

Second life battery use to support Electric Vehicles integration

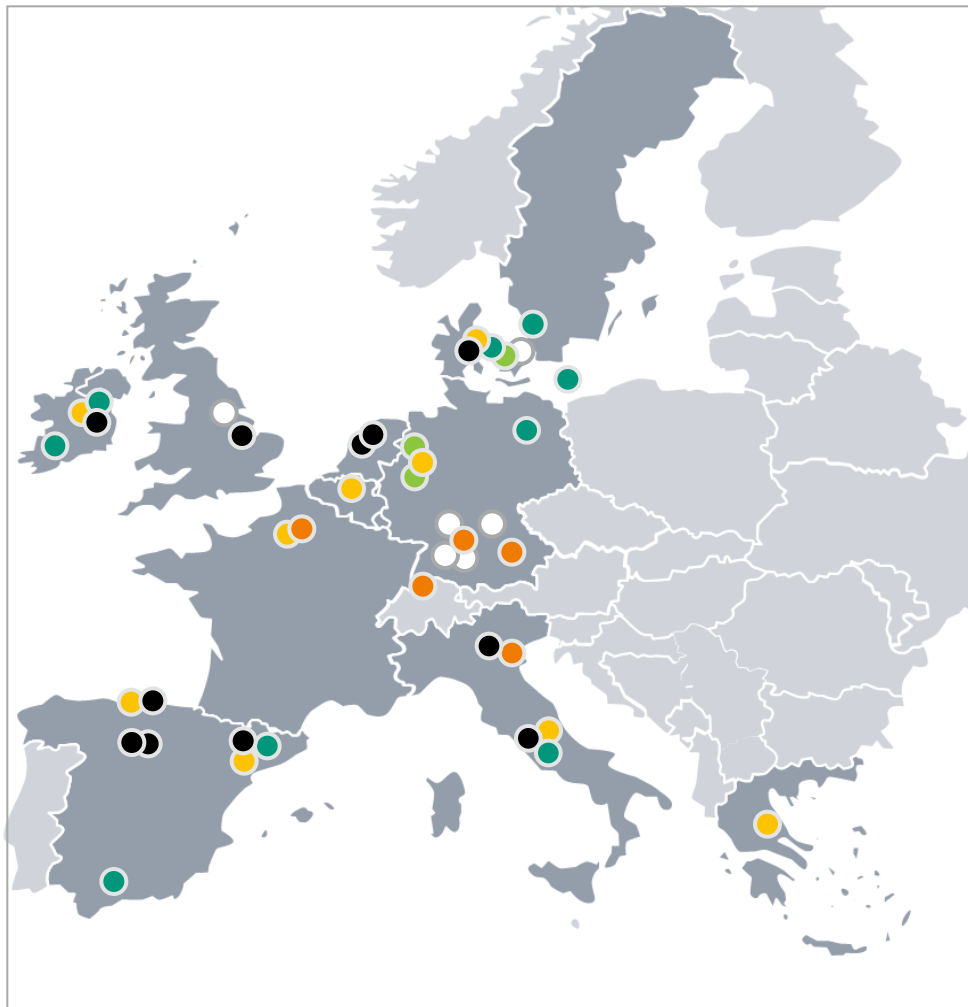
ENDESA

Barcelona, January 8, 2014

Narcís Vidal



Green eMotion with 43 Partners



○ Industries:

Alstom, Bosch, IBM, SAP, Siemens
(Project Coordinator)

● Utilities:

Danish Energy Association, EDF,
Endesa, Enel, ESB, Eurelectric,
Iberdrola, RWE, PPC

● Electric Vehicle Manufacturers:

BMW, Daimler, Micro-Vett,
Nissan, Renault

● Municipalities:

Barcelona, Berlin, Bornholm, Copenhagen,
Cork, Dublin, Malaga, Malmö, Rome

● Research Institutions and Universities:

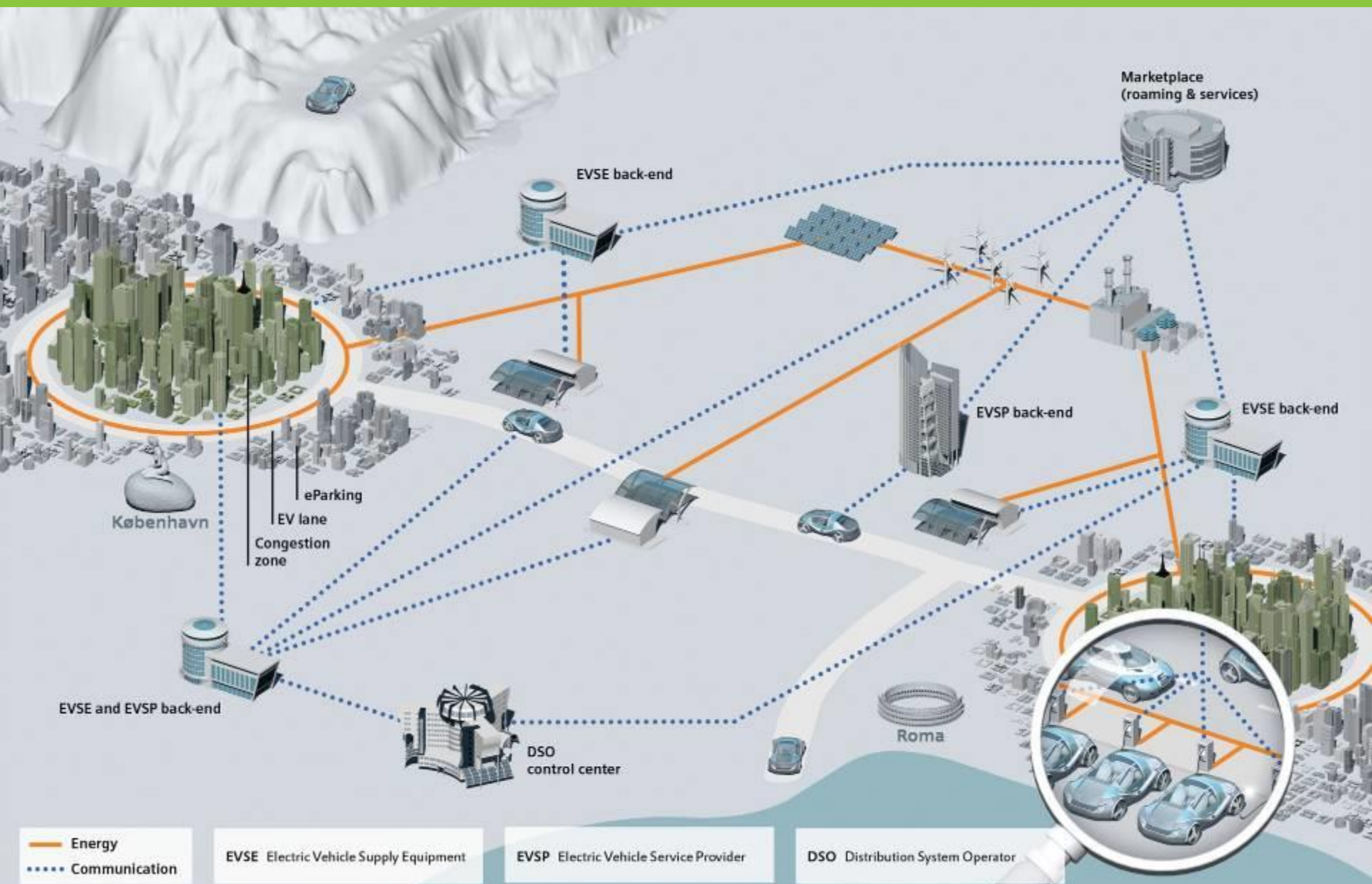
Cartif, Cidaut, CTL, DTU, ECN, Imperial,
IREC, RSE, TCD, Tecnalia, TNO

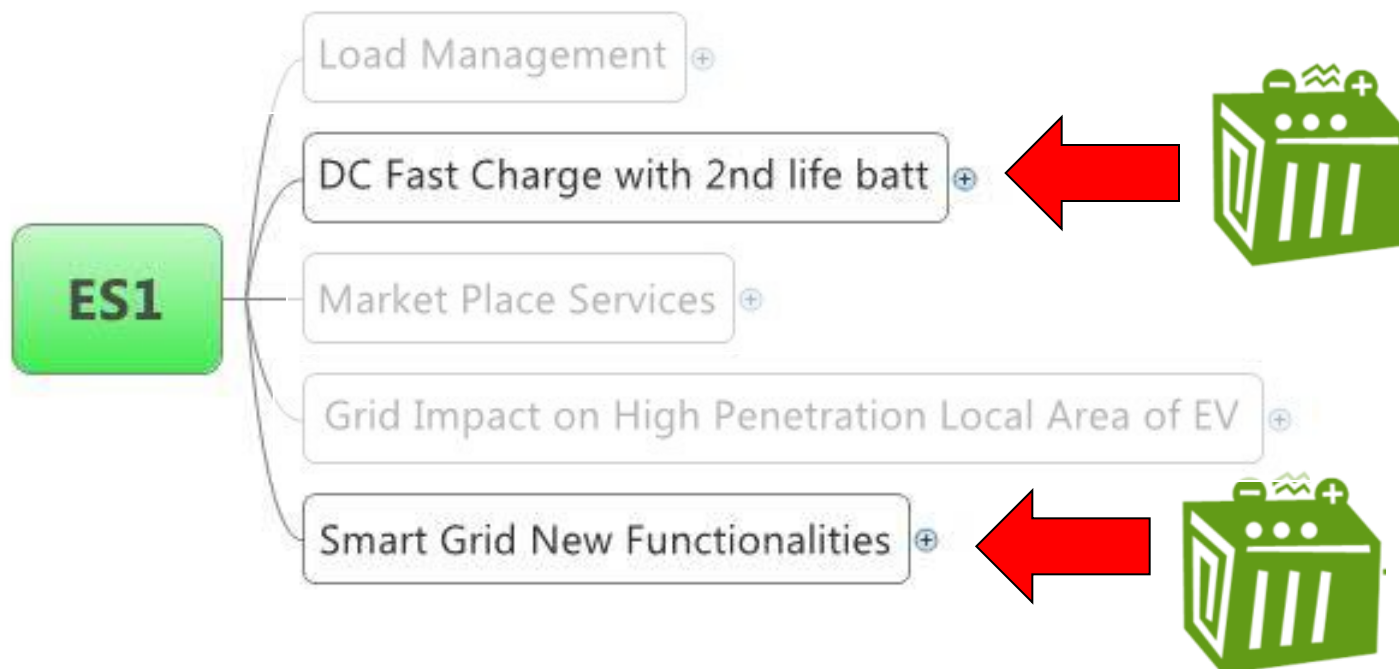


● EV Technology Institutions:

DTI, FKA, TÜV Nord

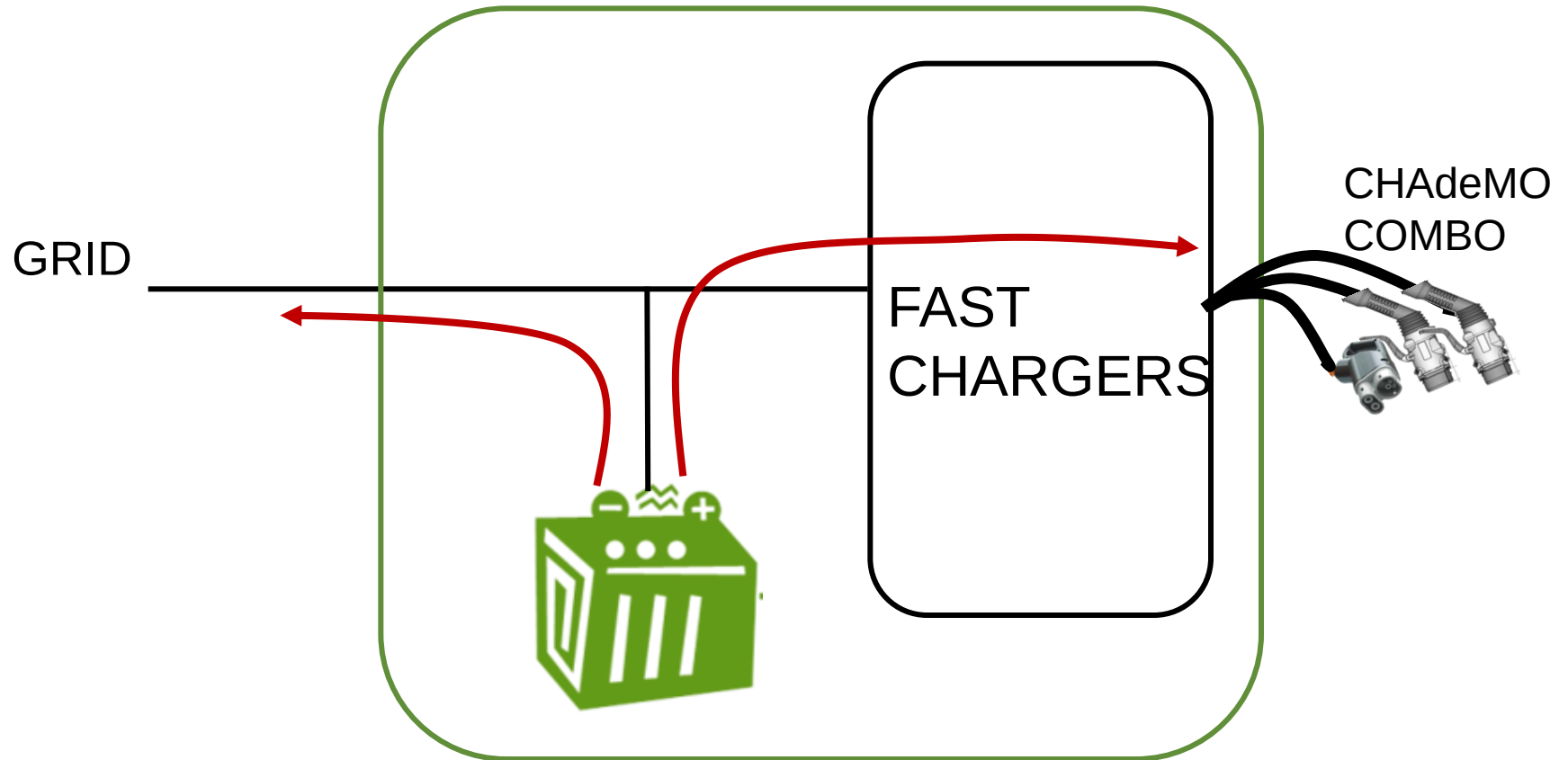
European electromobility – the big picture



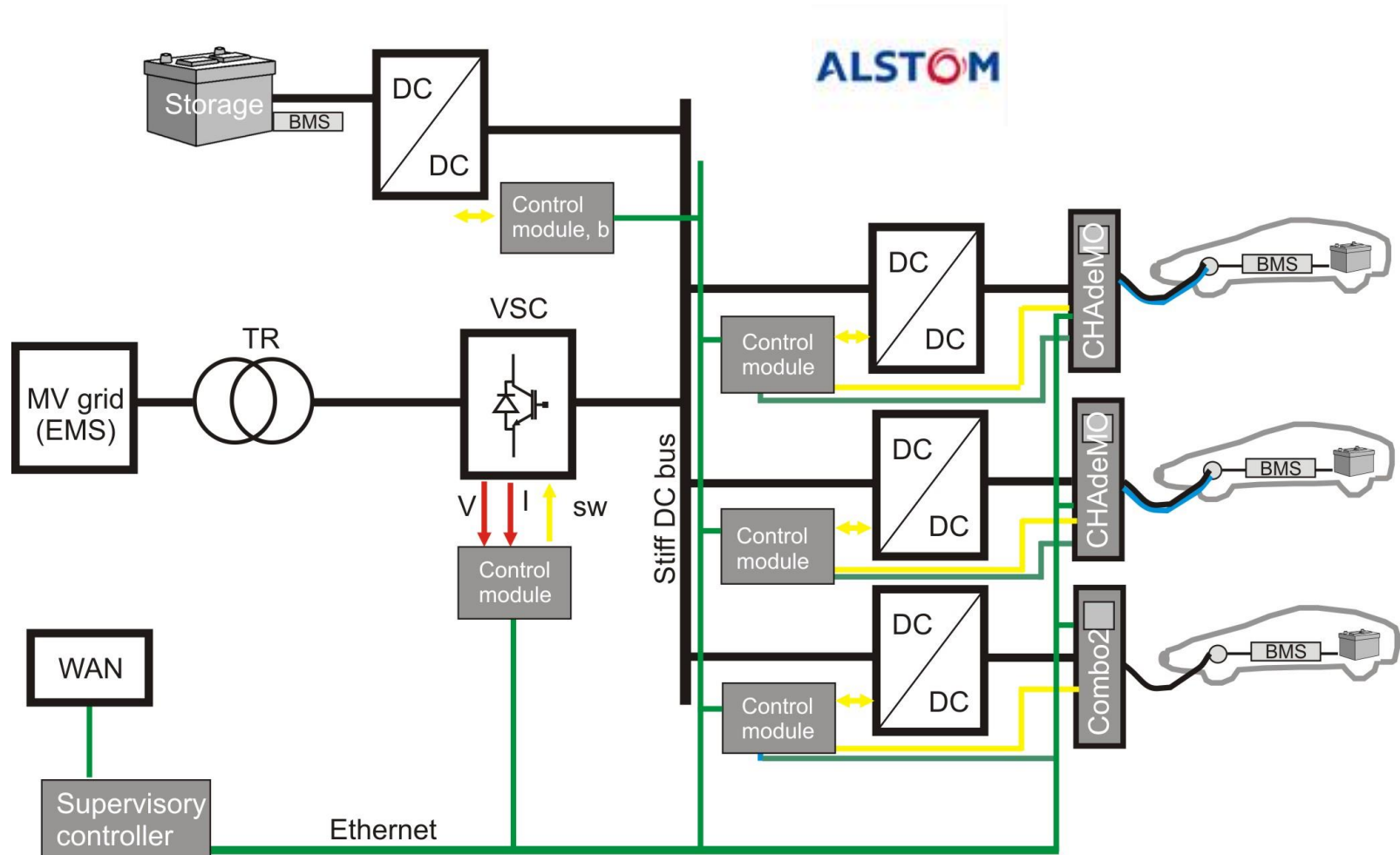




2. DC Fast Charge with batteries



2. DC Fast Charge with batteries

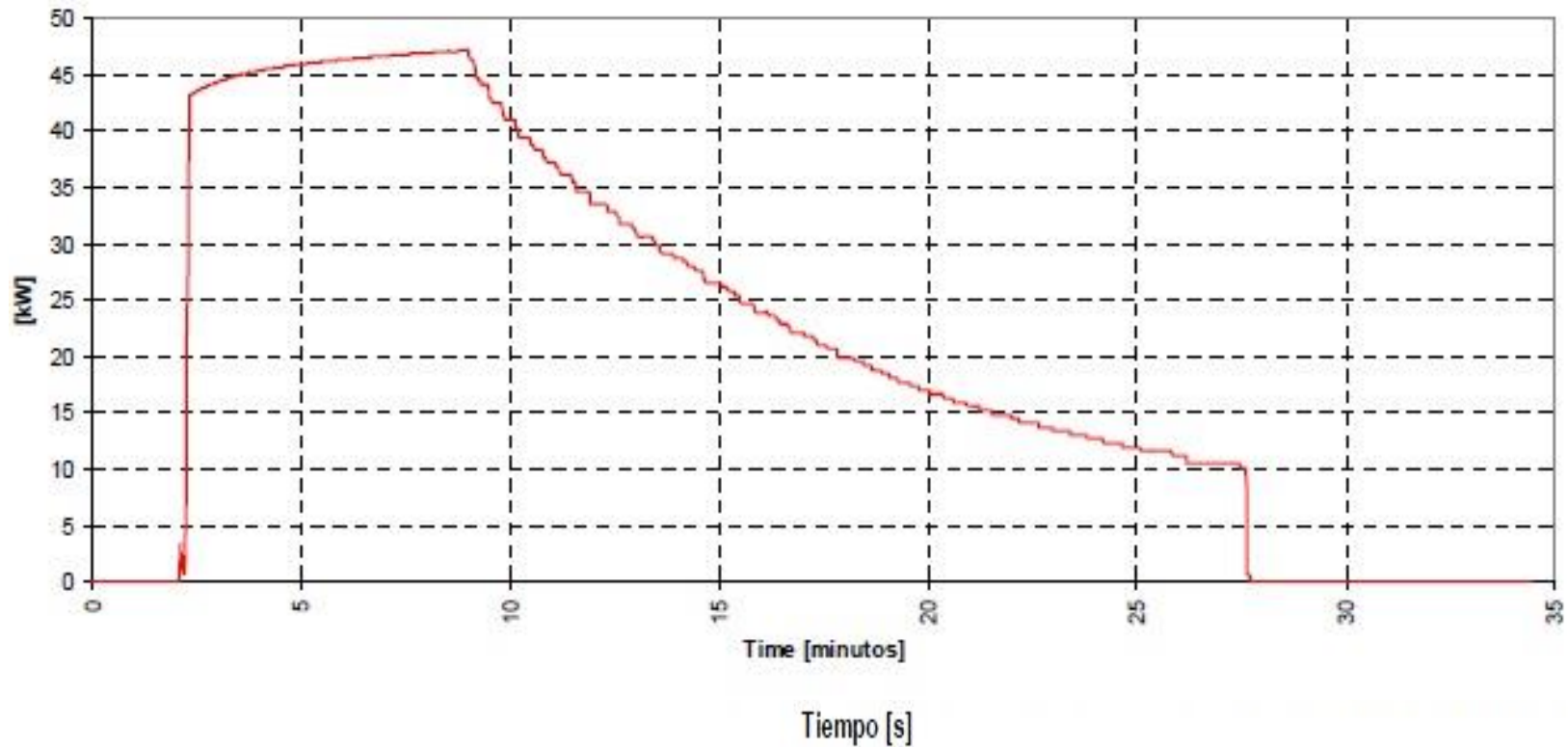


* signal lines are only indicative

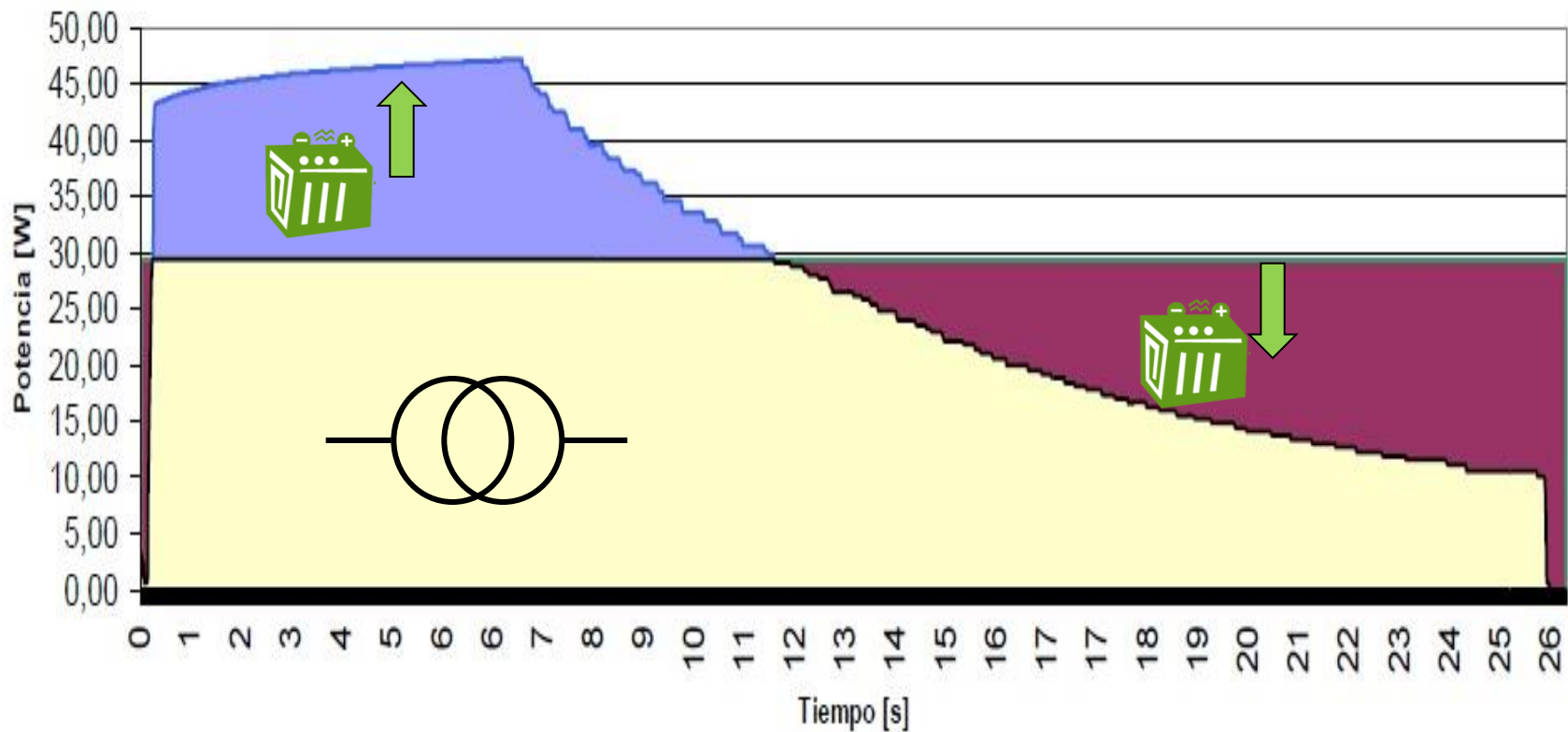
2. DC Fast Charge with batteries

		Fast Charge Malaga
Funcionalidad	Application	
Time Shift	Energy Arbitrage (Cost Minimization)	
	CO ₂ Emission Minimization	
	Peak Shaving	X
Power Quality	Instantaneous DSO Power Balancing	
	Islanded Mode	
	Reactive Power Balancing	X

2. DC Fast Charge with batteries

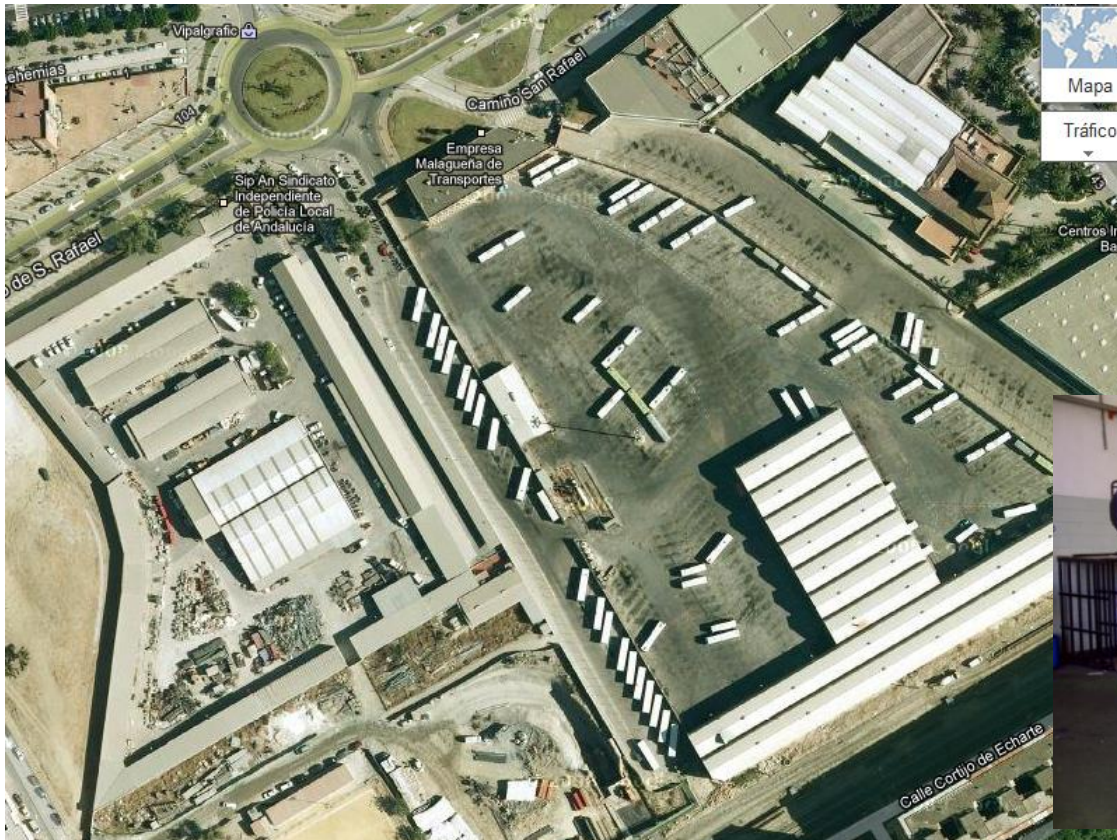


2. DC Fast Charge with batteries



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Location: Malaga Municipality Bus depot



2. DC Fast Charge with batteries



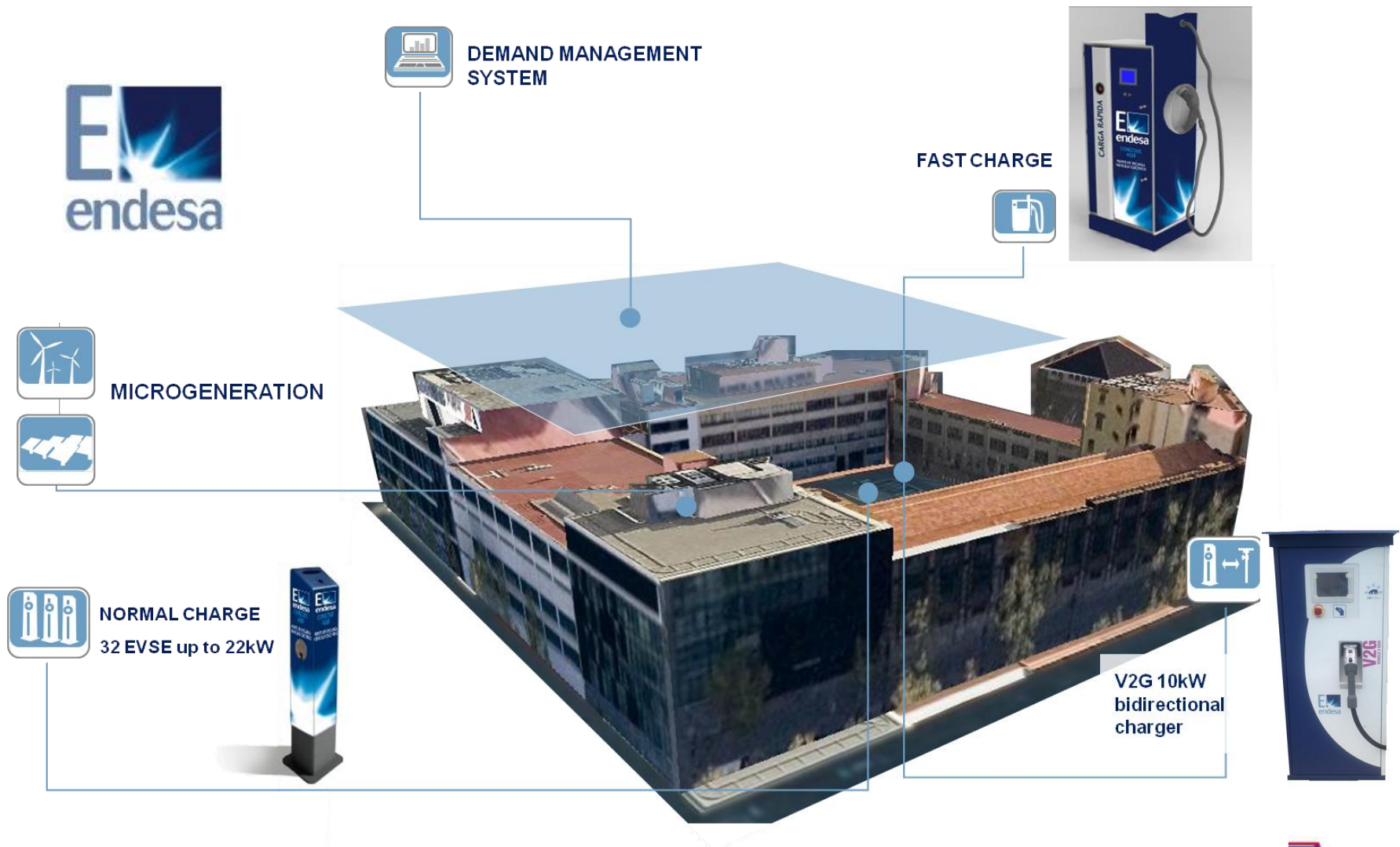




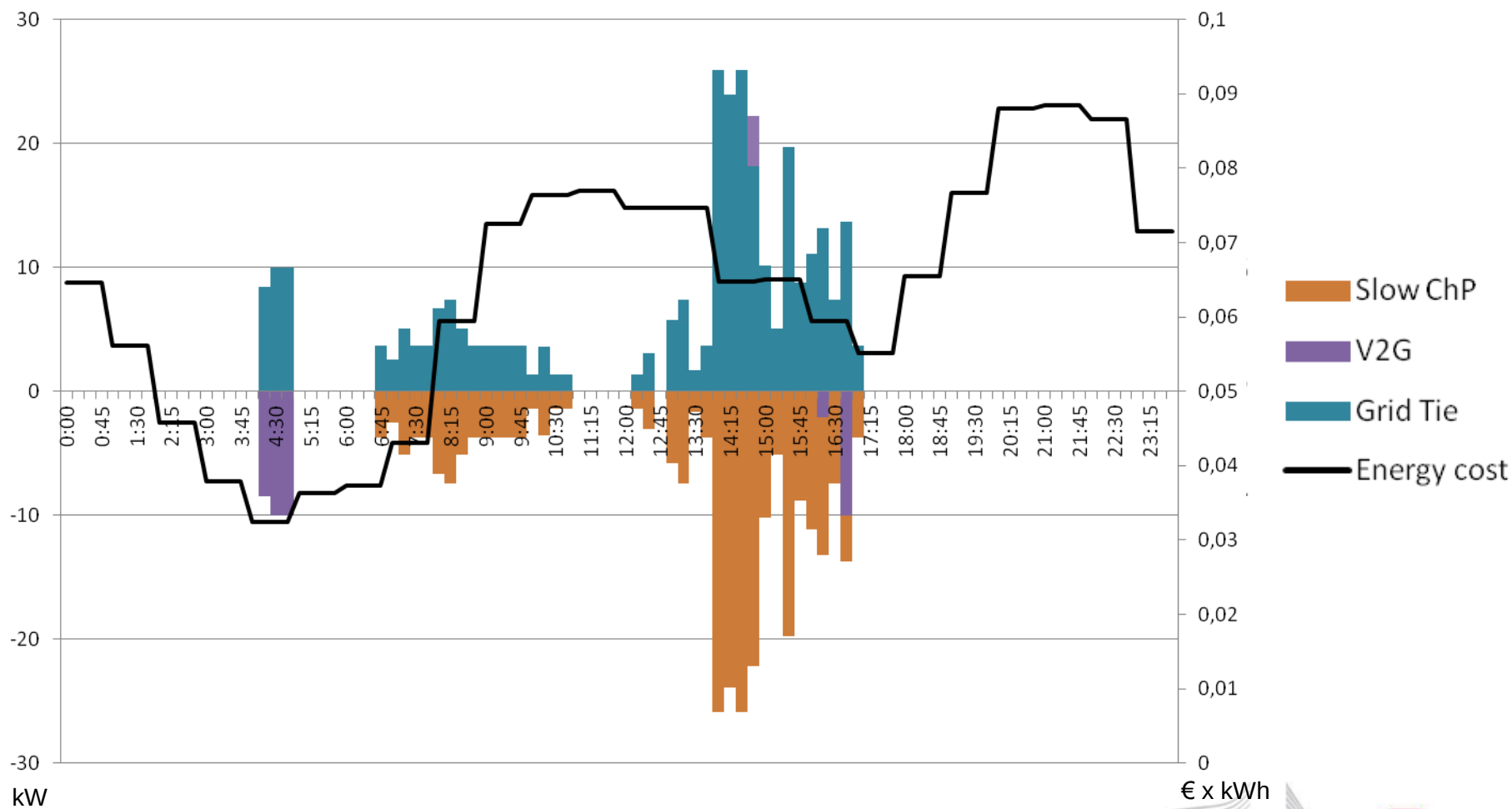
**2nd life batt +
40kW converter +
50kW Charger**



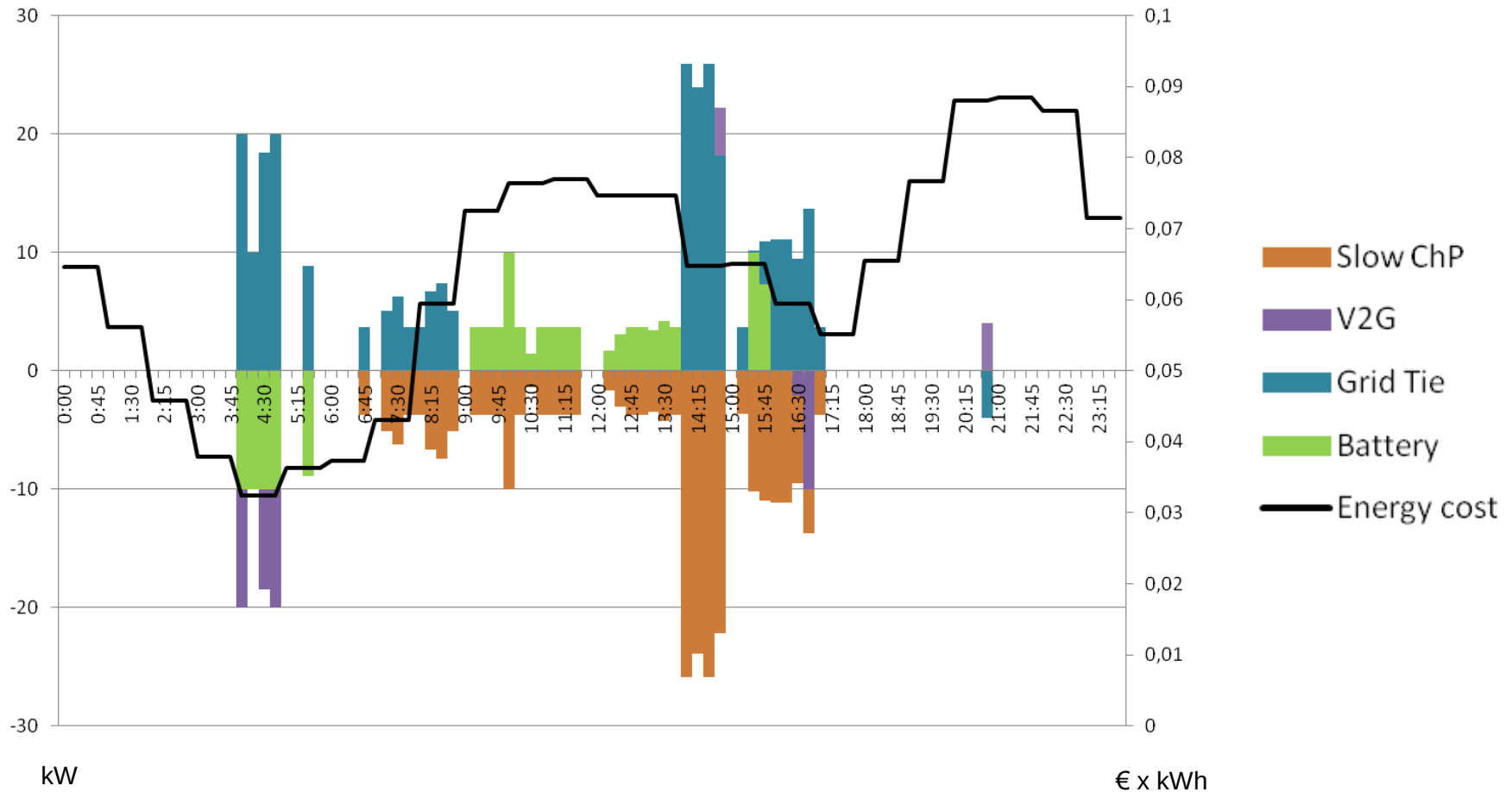
Grid Impact on Local High Penetration Area of EV



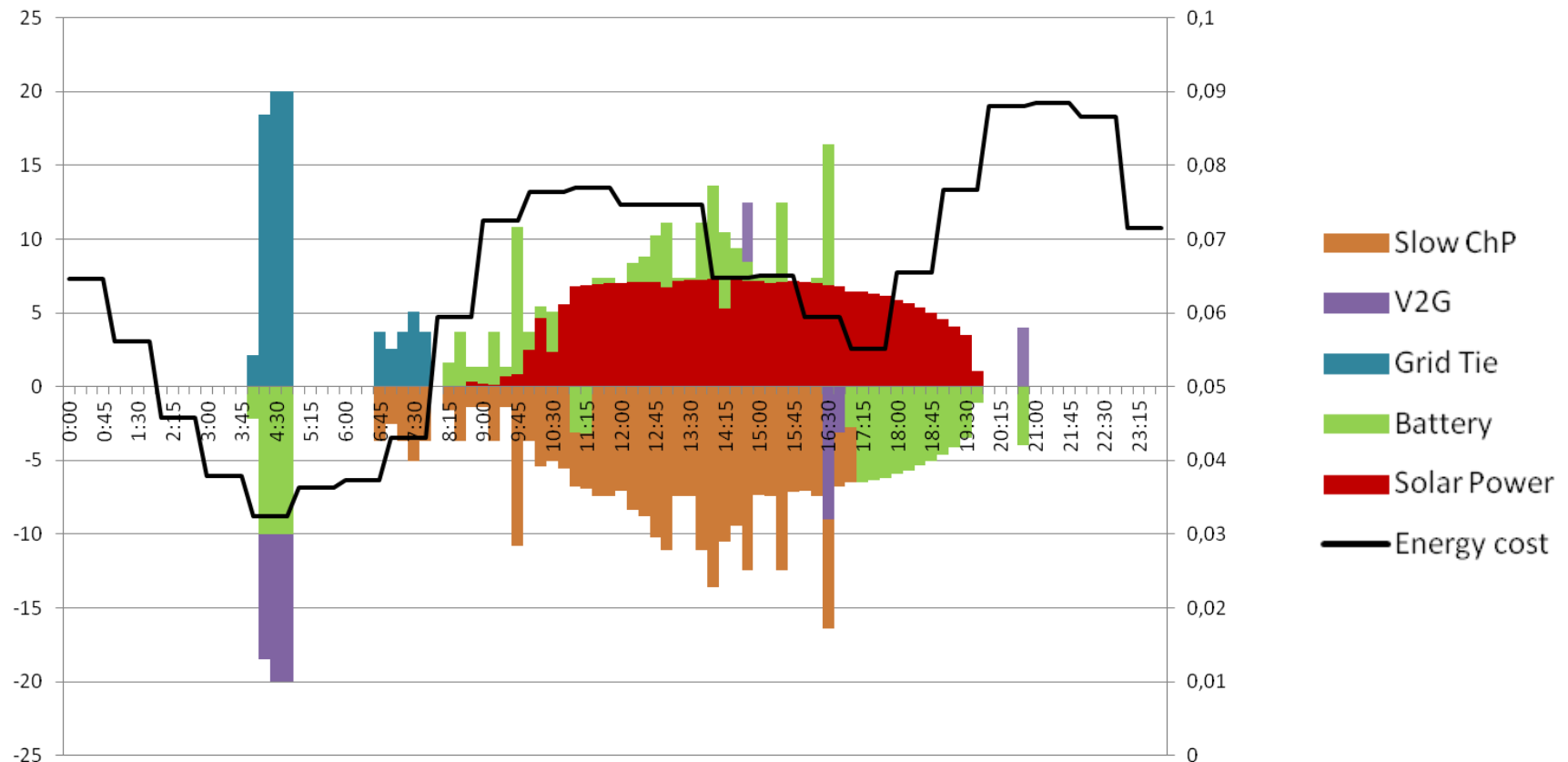
Smart Grid Management Schemes New Functionalities



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Smart Grid Management Schemes New Functionalities



Thank you for your attention.

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